

Carlos Narvaez

Embedded Systems Developer / Engineer

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PROFESSIONAL SUMMARY

Embedded instrumentation systems integrator with 10+ years of experience in electronics and software design, executing all embedded development stages: system conception, firmware design, programming and testing, hardware prototyping, and final PCB design and validation. Experienced in bioelectronics project development, general instrumentation design, and testing.

SKILLS

Firmware & Software: Embedded C, C++, Python, ASM, RTOS, Embedded Linux, Multithreading, Git, Debugging

Platforms: STM32, ESP32, ATmega, Microchip PIC, Raspberry Pi, BeagleBone Black, NVIDIA Jetson, Arduino

Hardware: KiCad, Altium Designer, PCB layout, SMD soldering, analog circuit design (AD620, INA114, MAX4194/4094), mixed-signal systems, BOM management, wire harnesses

Protocols & Lab: I2C, SPI, UART/Serial, CAN, One-Wire, MQTT; oscilloscopes, logic analyzers, digital multimeters

Soft skills: Problem solving, time management, teamwork, creativity, adaptability

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer

Oct 2025 – Present

Morphace (Self-Employed Contractor) · Greater Toronto Area

- Designed embedded circuits and PCBs in KiCad for skin-health and bioelectronics applications.
- Supported hardware validation and regulatory compliance under CSA SPE-3000:19.

Embedded Systems Developer

Sep 2025 – Present

Moon Trades (Self-Employed Contractor) · Toronto

- Leading sensor integration and embedded system architecture for the PEBBLE™ autonomous exploration platform.
- Selecting and integrating advanced instrumentation, including LiDAR, hyperspectral imaging, and XRF geochemical analysis systems.
- Developing a real-time processing architecture on NVIDIA Jetson Orin for GPS-denied navigation and edge AI; executing hardware integration for OCI Discovery Project underground validation.

Embedded Systems Engineer

Aug 2025 – Present

Robbie Insurtech (Self-Employed Contractor) · Mississauga, ON

- Evaluate and document new hardware sensor upgrades; support software and hardware integration of hard-tech solutions.
- Validated hardware integration using Linux-based tools; field-tested moisture sensors for performance and reliability.

Founder & Principal Engineer

Mar 2024 – Present

Rabbit Systems Inc. · Greater Toronto & Hamilton Area

- Designed a Command Line Interface library for STM32 and ESP32 microcontrollers.
- Designed a Collision Avoidance System prototype for racing cars using a Texas Instruments mmWave radar and NVIDIA Jetson.
- Designed and tested a low-power datalogger system for general-purpose applications.
- Execute requirements from verbal description to functional plan documentation, component selection, and layout and programming of functional PCBs using surface-mount soldering.

Instrumentation & Embedded Systems Developer

Nov 2020 – Mar 2024

Minard (Electronic Components Supplier) · Toronto

- Built firmware to simulate a TRIAXYS wave sensor for remote datalogger testing.
- Created a software architecture for a general-purpose microcontroller command line interface binding serial ASCII commands to MCU functions.
- Conducted system architecture planning, schematic design, PCB layout, and firmware development; integrated an OEM photoplethysmography sensor with a Java GUI.
- Prototyped a 3D-printed harmonic drive for low-cost robotics using a brushless motor.
- Developed a data logger prototype for maritime deep-sensing of seawater conductivity.
- Designed a wireless low-power electrocardiography acquisition device for laboratory rats.

Electronic Instrumentation Engineer

Feb 2018 – Oct 2020

National Autonomous University of Mexico (UNAM)

- Led and collaborated with mechanical and engineering teams developing instrumentation for astronomical observations.
- Maintained the robotic telescopes COATLI and DDOTI, their joint electronic systems, and main computers.
- Designed a CCD testing device using ATmega328 and laser LEDs; produced technical reports.
- Designed a meteorological measurement system for telescope domes using a digital anemometer and BeagleBone Black.

Electrical & Electronic Engineering Professor

Jan 2017 – Jan 2018

National Technology Institute of Mexico (TecNM)

- Facilitated courses in Advanced Programming, Instrumentation, and Real-Time Programming; tutored students on embedded systems.

EDUCATION & CERTIFICATES

Computer Programmer Diploma, Seneca Polytechnic 2023

Master's in Electronic Engineering (Bioelectronics), Autonomous University of San Luis Potosi 2016

BEng, Mechatronics Engineering, Technological Institute of San Luis Potosi 2013

Advanced Multilayer PCB Design with Altium Designer · Basic PCB Design with Altium · STM32 HAL API Driver Development

LANGUAGES & INTERESTS

Languages: English (professional working) · Spanish (native)

Interests: Clay sculpting · Urban cycling · Homemade electronic projects · Hiking · Exploring the city